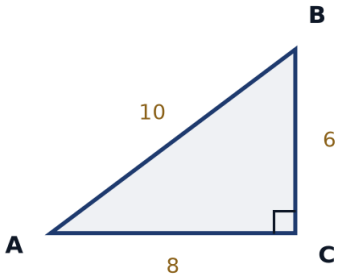
Angles of polygons

Core Concepts: Angles of polygons

1 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 104°

b. 114°

c. 96°

d. 149°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 13

c. 11

d. 13

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97°

b. 107°

c. 89°

d. 149°

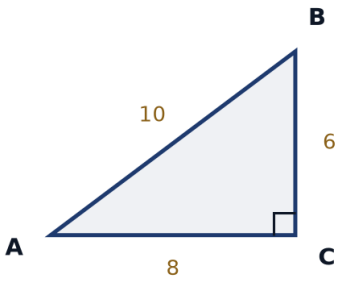
5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm

c. 20 cm

d. 45 cm



6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 11

b. 15

c. 13

d. 15

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 149°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

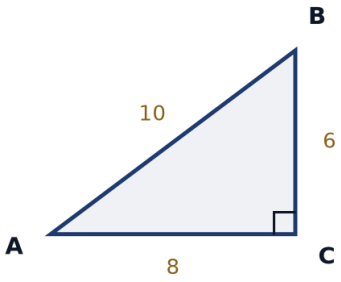
9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 13

b. 17

c. 15

d. 17



Exam-Style Problem Solving & Modeling

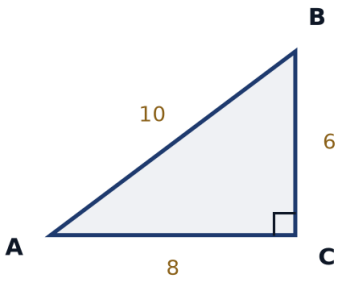
10 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 149°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 15

b. 19

c. 17

d. 19

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